

Sajid Ahmed

AI / Machine Learning Engineer · Data Scientist

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PROFILE

AI engineer who puts grounded LLM systems into production, not prototypes. Shipped Copilot agents now live across nine countries, on a governed semantic layer I took from zero to production and built to be machine-readable. Five years at Vodafone Group building the data foundations and automation that 14 national operating companies ("markets") run on: €19.5k/year of vendor spend removed, ~390 analyst hours a year returned per market, and the numbers reported to senior leadership made defensible.

EDUCATION

University of Cambridge — MPhil, Economics and Data Science (part-time) · Sep 2025 – Jul 2027

Year 1 complete: 70, 70, 73, 76 — highest marks in Seminar Series and Group Presentations. Econometrics, causal inference and unstructured data; applied vision LLMs and NLP to real extraction problems, with coursework modelling built on Vodafone channel data (approved, no personal data). Self-funded; dissertation in progress.

Queen Mary University of London — BSc (Hons) Economics, Mathematics and Statistics · First Class, 2020 · Numerical Computing with C++ (99%)

TECHNICAL SKILLS

- **GenAI & LLM engineering:** RAG (hybrid sparse + dense retrieval, reciprocal rank fusion), agent grounding, evaluation and hallucination suppression, vision LLMs, prompt engineering, LangChain, Hugging Face Transformers, Open Knowledge Format (OKF)
- **ML & statistics:** supervised regression and classification, anomaly detection, forecasting, econometrics, A/B testing, scikit-learn, model evaluation and error analysis
- **Programming & engineering:** Python (pandas, NumPy, scikit-learn), SQL (BigQuery, MSSQL), R, Git, REST APIs, Airflow, Cloud Composer, GCP, Docker, CI/CD, ETL pipelines, Streamlit
- **Modelling, governance & BI:** semantic layers, LookML, ontology design, metric standardisation, hub-and-spoke governance, Looker, Tableau, Power Platform, Dynatrace (administrator), synthetic and real-user monitoring

EXPERIENCE

Channels Data and AI Specialist — Vodafone Group, London · Aug 2026 – Present

Channels & Products Analytics, AI & Data Tower.

- Designed, built and shipped **Microsoft Copilot agents**, live July 2026 with **15+ users across nine countries**. Grounded on a market-segmented knowledge base authored in Google's **Open Knowledge Format**, chosen because tightly scoped sources narrow the model's scope to hallucinate — the difference between a demo and something safe to use in incident response. Cut specialist questions routed to me from ~10 a week to 1–2.
- Built and evaluated an **LLM reporting agent** across 14 markets, diagnosing 12+ distinct failure modes through structured testing and eliminating the core bug class via a 9,500-formula, zero-error data pipeline.

Channels Insight Analyst — Vodafone Group, London · Aug 2023 – Aug 2026

Channels & Products Analytics, AI & Data Tower. Supporting 14 markets.

Production GenAI and agent engineering

- Made the semantic layer **agent-readable and LLM-safe**: an OKF-based agent framework extended with authorship tracking and an agentic pipeline auto-publishing schema changes to Confluence and GitHub, plus query guardrails preventing runaway compute cost on a multi-terabyte dataset.
- Shaped the market-facing use cases for the Group's conversational-analytics capability and **demoed it to senior leadership**; it became the first use case taken forward under the Group agentic AI programme.

Data modelling and governance

- Took a governed semantic layer in **Looker/LookML from zero to a production MVP** on live market data, as the sole expert on a tool nobody in the function had used, delivered two weeks ahead of a hard deadline. Sole **LookML SME** across markets; authored parallel technical and stakeholder documentation and trained colleagues to operate independently.
- Designed the **hub-and-spoke governance model** now adopted as the Group standard — a governed central core with controlled local extension — by framing three costed options and their trade-offs for the product lead rather than presenting a single answer.

Analytics, measurement and automation

- Designed the **Impact Score**, replacing an outlier-sensitive responsiveness average with a measure weighted by customers affected. Live in **14 markets since January 2025**; front-end teams now prioritise their backlogs by it. Surfaced a homepage improving from 6.34s to 2.59s for 280,000 monthly customers — the largest gain on the estate, invisible in the previous aggregate. Also specified Vodafone's **digital**

carbon-footprint KPI on the Sustainable Web Design standard, which another team rolled to all 14 markets from my spec.

- **Instrumented 3.3M weekly personalisation requests** between Pega and the MyVodafone app, decomposing the journey to attribute a ~12% failure rate to specific stages. Isolated three simultaneous points of customer loss owned by three different teams — precisely why none could see it.
- Automated a market's weekly operational reporting end to end via the **BigQuery API in Python: 7–8 hours a week to ~30 seconds** (~390 hours a year per market), now sponsored by the Head of IT & Digital to scale from local POC to a cross-market cloud service. Separately replaced a **€40k two-year vendor contract** for app-store review data with an in-house pipeline across three markets, ingesting 15,600+ reviews for under €500 a year — **~€19.5k/year net saving**.

Platform ownership

- **Sole administrator and SME for Dynatrace** across markets — the reference point for the 4–5 major incidents arising every couple of months, and accountable for tenant and access management after removing an external vendor dependency.
- Delivered an **end-to-end exclusions workflow** across the Dynatrace API, Power Automate, SharePoint, PowerApps, Teams and Python. **Adopted by ~27 people across 10 markets**, processing ~2,000 failures a week, returning ~2 hours of specialist time daily and cutting month-end reporting from 2–3 days to half a day for two people (~48 person-days a year). Roughly one request in five is rejected, which is what makes the resulting figures defensible to senior management.

Technology Graduate — Vodafone Group, Newbury/London · Sep 2021 – Aug 2023

Rotations across Channels & Products Analytics, Internal Audit Analytics and Business Insights. Applied **anomaly detection to reduce API errors by 50%**, built a **BigQuery pipeline handling 1TB+** in real time, and designed an **Airflow/Cloud Composer CI/CD framework** for automated ML model training.

PROJECTS

- **RAG question answering over a scanned archive** (*self-directed, 2026*) — vision-LLM extraction of 30 scanned pages of 1880s–1920s literature into structured JSON, then hybrid retrieval (TF-IDF + dense embeddings via reciprocal rank fusion), with generation constrained to refuse outside the corpus. The evaluation is the interesting part: a first eval set saturated at 100% and could not catch a regression, so I rebuilt the metrics and documented the remaining gaps rather than claiming production readiness.
- **F1 race-strategy optimisation** (*Udacity Data Scientist capstone, 2024*) — supervised regression predicting optimal pit-stop lap and stop count from tyre degradation, position change and race pace; scikit-learn

ensembles tuned and compared on MSE and R^2 , benchmarked against teams' actual in-race strategies. Also built a multi-label NLP disaster-response classifier served via Flask.

- **AI image generation** (*Vodafone internal hackathon, team lead, 2024*) — led a team of four from concept to a working diffusion-model prototype in 48 hours, using bounding-box pre-grounding to control asset size, position and orientation. Runner-up, Vodafone TechBoost innovation competition (2023).

RECOGNITION & WRITING

- **Five Vodafone Star Awards across four consecutive years (2022–2025)**, three nominated by the Principal Manager for Channels Performance & Analytics — cited for *"exceptional ownership... tangible business impact... directly supporting key CTO priorities"* (2025); includes two *Create the Future* awards for mentoring.
- **Writing:** publish on AI and agentic workflows at medium.com/@sajid.ahmed, including *"The New Dictionary for AI and Agentic Workflows: OKF in LookML"* and *"The Three Killers to Any Data Science Project"*.
- **Certifications:** LangChain Academy, Foundation: Introduction to LangChain in Python (2025) · Udacity Data Scientist Nanodegree (2024) · Microsoft Certified: Azure Data Fundamentals, DP-900 (2023).